Fooling Aerial Detectors by Background Attack via Dual-Adversarial-Induced Error Identification

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***Abstract*— Recent developments in adversarial attack have witnessed the success of background attack against object detectors. However, most existing methods attack detectors by luring targets into background. Therefore, an innovative background attack framework via dual-adversarial-induced error identification (BADEI) is proposed to attack detectors by deceiving background as targets, as well as deceiving targets as background, where the attack performance can be greatly enhanced by these two kinds of induced error identification. Specifically, a mechanism that generates the adversarial background is proposed to result in dual error detection, where the background can conceal the specified targets and cause the misclassification of the adversarial pattern in the background as a specific category. Moreover, an unoccluded training strategy (UTS) that leverages the target mask of an image is introduced to strategically place adaptive adversarial background beneath the targets while optimizing and updating the pixel values of the background outside the target region, which can enhance attack effectiveness for adversarial background, significantly degrade the targets' average accuracy, and enhance the robustness of background. Finally, a dual deceptive loss function (D^2LF) is carefully formulated to generate false negatives (FNs) and false positives (FPs) to achieve untargeted attacks for hiding objects as well as targeted attacks for erroneously recognizing objects. Extensive experiments and comparative analysis of various victim network models on two datasets (including the dataset for object detection in aerial images (DOTA) and RSOD dataset) confirm that the proposed framework exhibits superior performance over the state-of-the-art methods in both digital and physical scenarios.**

***Index Terms*— Aerial detection, background attack, dual adversarial induction, false negatives (FNs), false positives (FPs).**

I. Introduction

IN RECENT years, deep neural networks (DNNs) have found extensive application across various domains, such as image classification [1], [2], [3], [4], [5], [6], [7], object detection [8], [9], [10], [11], [12], semantic segmentation [13], video retrieval [14], face recognition [15], [16], and autonomous driving [17]. However, DNNs exhibit the characteristics of uncertainty, large parameter counts, nonlinearity, and scalability, resulting in DNNs undeniably presenting us with certain challenges. In addition, it has been confirmed that DNNs are vulnerable to adversarial examples [18], which are crafted by adding adversarial perturbations. These perturbations that are imperceptible to human can mislead DNNs into generating a different wrong prediction and even extend to the real world, which poses significant concerns to some safety-critical application scenarios. Consequently, research on adversarial attacks and malicious examples has increasingly provoked attention [19], [20], [21], [22]. Furthermore, the study on adversarial attacks helps to enhance the adversarial robustness and generalization ability of DNNs, as well as their interpretability and reliability.

Adversarial attacks can be divided into digital attacks and physical attacks according to the field of realization. Digital attack refers to the manipulation of digital pixels after camera imaging, where the pixel values of an image can be directly modified. Common digital attack methods, fast gradient sign method (FGSM) [23], Carlini and Wagner (C&W) [24], projected gradient descent (PGD) [25], momentum iterative FGSM (MI-FGSM) [26], and AutoAttack (AA) [27], primarily focus on image classification task of the digital domain. These common methods excel on DNN models but are challenging to implement in the real world. Moreover, digital perturbations are typically global and inconspicuous, making them difficult to be clearly captured by cameras. These drawbacks have prompted researchers to explore more practical attack methods in the physical world, the so-called physical attack. Physical attack refers to the manipulation of physical objects before camera imaging. Because detection systems work by scanning objects directly, previous physical attack methods mostly attack the targets in the form of color [28], makeup [29], laser [30], watermark [31], sticker [32], and texture [33], which change the appearance of objects and provide adversarial examples to intentionally induce incorrect prediction. However, these methods are often restricted. For example, the target categories of makeup and watermark attacks are more homogeneous and usually effective in white-box attacks, with limited transferability in black-box attacks [29], [31]. And due to the concavity of the objects themselves, these methods are inconvenient to implement [21]. In addition, when physically implementing these methods, brightness, viewing distances, and object scales, will result in poor attack effectiveness [34].

Although there are certain problems with adversarial attacks, object detection algorithms based on convolutional neural networks (CNNs) have achieved remarkable success in aerial detection [35], [36] in recent years. In addition,

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adversarial attacks on aerial detectors have also received extensive attention [12]. Most of the current work still involves modifying the targets themselves, i.e., altering the appearance of objects within physical scenes, to provide adversarial examples for aerial detection devices. Simultaneously, most aerial detection images feature small target scales and encounter complex physical conditions such as lighting and noise, and posing significant challenges. Some latest works [37], [38], [39], [40], introduce adversarial patches derived from general real-world scenarios. They elaborate a malicious patch to be placed on the target object or pasted elsewhere in the field of view, causing the detectors to make incorrect prediction. Despite their significant success in physical attacks, where adversarial patches are widely applied in diverse tasks, for example, aerial detection and facial recognition, they inevitably exhibit certain limitations. First, existing adversarial patches are predominantly foreground-based, emphasizing the obscuration of ground-level targets. This is often proved inconvenient and underutilized contextual background information. Therefore, the background attack method with perfect attack effect is an urgent problem to be studied. In addition, compared with natural images, achieving the transformation between the digital domain and the physical world requires consideration of complex environmental factors. At the same time, the improper setting of the patch training strategy usually leads to unsatisfactory and unstable attack effectiveness [21], [22]. Last but not least, current research focuses exclusively on single-induced attacks, i.e., attacking detectors by fooling targets into the background, which typically prioritize target concealment and exhibit limited capabilities for target confusion or misdirection against aerial detection.

To settle the above problems, a background attack approach is introduced by utilizing dual-adversarial-induced error identification (BADEI) to attack detectors by fooling background as targets, as well as fooling targets as background, which can conceal and misclassify targets, as shown in Fig. 1. Specifically, we aim to design a background adversarial attack framework that generates a background exhibiting strong adversarial robustness while adapting to the complicated background in aerial detection, encompassing variations in scale, lighting, noises, and other complexities. Moreover, an unoccluded training strategy (UTS) is put forward by placing the background directly beneath the targets with masks obtained from the instance extractor, which harnesses the background information within the image for the update and optimization of adversarial background. Then, a dual deceptive loss function (D^2LF) is proposed to enhance the flexibility of the mechanism, which can deal with both targeted and untargeted tasks. For a comprehensive evaluation, extensive robustness validation experiments are conducted based on diverse victim models, including CNN-based detectors, Transformer-based detectors, one-stage detectors, two-stage detectors, anchor-based detectors, and anchor-free detectors. The results reveal the vulnerability of DNNs and validate the effectiveness of the proposed BADEI in both the digital and physical domains.

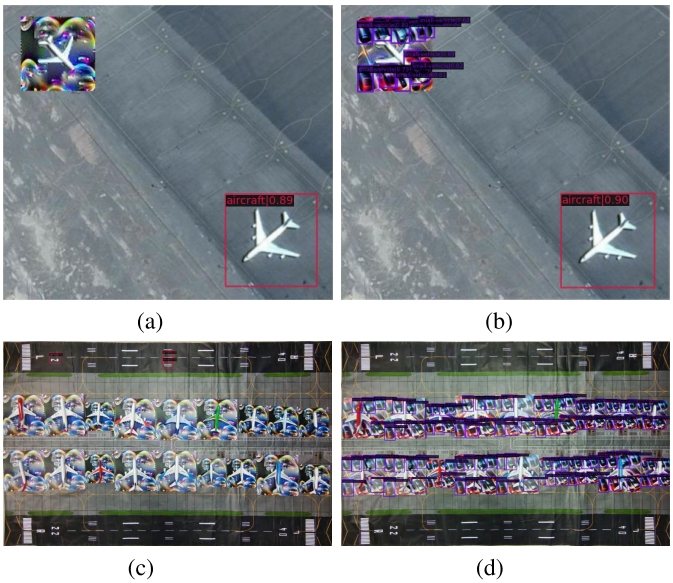


Fig. 1. Performance of adversarial attacks on aerial detection in the digital domain and the physical world. (a) and (c) FNs conceal targets from the detectors in the digital and the physical world. (b) and (d) FPs obscure targets and misclassify them as a specified category in the digital and the physical world.

In summary, the main contributions are fourfold as follows.

1) A background attack framework is designed to deceive aerial detectors by dual adversarial induction. The framework makes use of contextual information to design and generate an adversarial background, which demonstrate outstanding performance, regardless of whether it is a white-box or black-box setting, or in the digital or physical domain.
2) An UTS is developed to craft adversarial perturbations. The adversarial backgrounds are placed under the targets with masks and inverse masks, which can effectively conceal the objects from detection and are easy to implement physically.
3) A D^2LF is proposed, simultaneously elaborating false negatives (FNs) and FPs against aerial detection, both of which demonstrate the flexibility of the mechanism and exhibit excellent attack effectiveness.
4) To the best of the authors' knowledge, this article is the first to make efforts in conducting benchmark tests for dual-adversarial-induced error identification against aerial detection. Rigorous and comprehensive experiments are implemented in both the digital and physical world to validate the attack effectiveness and transferability of adversarial background, thereby confirming the adversarial robustness of different detectors.

The rest of this article is organized as follows. Section II briefly reviews the related work of adversarial attacks. In Section III, a detailed exposition of the proposed BADEI framework is provided. Then, the efficacy of the generated adversarial background is substantiated through adequate validation in Section IV. Finally, Section V summarizes the introduced BADEI approach and delves into the discussion and prospects of dual-adversarial-induced background attacks.

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II. Background and Related Works

This section first briefly introduces background and related works on digital adversarial attacks and physical adversarial attacks. Subsequently, the development of adversarial attacks against aerial detection is discussed.

A. Digital Adversarial Attacks

The vulnerability of DNNs to adversarial samples has garnered significant attention. The study of adversarial phenomena begins with image classification tasks in the digital domain [41]. Generally speaking, digital adversarial attacks are mainly divided into white-box attacks and black-box attacks.

According to the production mechanism, white-box digital adversarial attacks can be divided into optimization-based attacks, gradient-based attacks, and hyperplane-based attacks. Representative attacks based on optimization include limited memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) [41] and C&W [24]. By finding the minimum loss function additive perturbation term leading to DNNs' misclassification, L-BFGS transforms the problem into a convex optimization problem to find adversarial perturbations. C&W maximizes the margin between logits by L_2 or L_∞ regularization on the input, thus expanding the search space for the optimal solution. Classical gradient-based adversarial attack algorithms include FGSM [23] and PGD [25]. FGSM applies a single-step perturbation to the test image, where the perturbation direction is opposite to the gradient of the loss function. PGD initializes perturbations from random points within a norm ball near the test image and introduces gradient projection rather than gradient clipping. DeepFool [42] relies on hyperplane, which uses the L_2 norm to calculate the minimum distance between a normal sample and the classification boundary of the target model to generate adversarial perturbations for nontargeted attacks.

Black-box attack research is more practical and can be categorized according to the generation approach, including transfer-based attacks, confidence score query-based attacks, and decision boundary-based attacks. Local substitute model attack (LSMA) [43], zeroth-order optimization (ZOO) [44], and boundary attack [45] are one of the classical algorithms in adversarial attacks, respectively. LSMA approximates the decision boundary of the target model by creating a substitute model and produces adversarial examples based on the current substitute model. ZOO employs the zeroth-order optimization method, which utilizes the symmetric difference quotient to estimate gradients and Hessian matrices. Boundary attack relies solely on the category labels obtained from the output of the classification model without requiring gradients or confidence scores.

However, the perturbations generated by digital attacks are often imperceptible, making it difficult for cameras to capture these perturbations. Hence, it is challenging to apply digital attack methods to the physical world.

B. Physical Adversarial Attacks

Adversarial examples have been shown to generalize to the real world, including autonomous driving, face recognition, and security monitoring. Kurakin et al. [46] first introduce physical attacks, who discover that DNNs are susceptible and vulnerable to adversarial attacks in real-world scenarios. Sharif et al. [47] generate an adversarial eyeglass frame based on facial biometric systems to impersonate an identity or evade the face recognition system. Zhou et al. [48] develop a cap with LED lights to counter face recognition systems, which is realized by illuminating the subject using infrared according to the adversarial examples. Face recognition systems can be bypassed or deceived, while the infrared perturbations remain imperceptible to raw eyes through this method. In the context of automatic driving, Eykholt et al. [49] utilized images of traffic signs under different perspectives and distances to identify vulnerable regions. They successfully attacked traffic sign classification networks to fool you only look once (YOLO) [50] by placing black-and-white stickers on these regions. Tu et al. [17] generate universal 3-D adversarial objects placed on the rooftop of any target vehicle to fool light detection and ranging (LiDAR) detectors and hide the vehicle entirely. Thys et al. [40] are the first to perform a white-box attack against the single-stage detector [50], where they print the adversarial pattern onto a piece of cardboard and then hold it in hand to evade pedestrian detection. Wu et al. [51] consider the deformation of the nonrigid body before generating adversarial patches on clothes and they benchmark the effectiveness of adversarial patches under both white-box and black-box settings, and quantify transferability of attacks among datasets, object classes, and detector models.

C. Adversarial Attacks Against Aerial Detection

Currently, most research on adversarial attacks primarily focuses on natural images, with relatively limited studies on adversarial attacks in satellite and aerial images. Czaja et al. [52] and den Hollander et al. [53] conduct research on adversarial attacks specifically for aerial imagery classification and detection in the digital world, but they have not deployed them in real-world scenarios. Du et al. [54] demonstrate one of the first efforts on physical adversarial attacks on aerial imagery, whereby adversarial patches are optimized, fabricated, and installed on or near target objects (cars) to significantly reduce the efficacy of the object detector. Lian et al. [21] propose a novel adaptive-patch-based physical attack (AP-PA) to evade aerial detectors, in which the physical varying factors and different object scales are properly accommodated. Besides, they confirm the effectiveness of adversarial patches in dynamic physical circumstances. Then, a contextual background attack (CBA) [22] based on AP-PA is proposed, where contextual background adversarial patches are gifted with solid fooling efficacy against object detectors. However, most of the above-mentioned methods are often limited to single-induced attacks, and their effectiveness lacks universality in both digital and physical domains.

III. Methodology

In this section, we commence by formulating the problem and subsequently provide an in-depth explanation of our proposed approach, denoted by BADEI, which specifically

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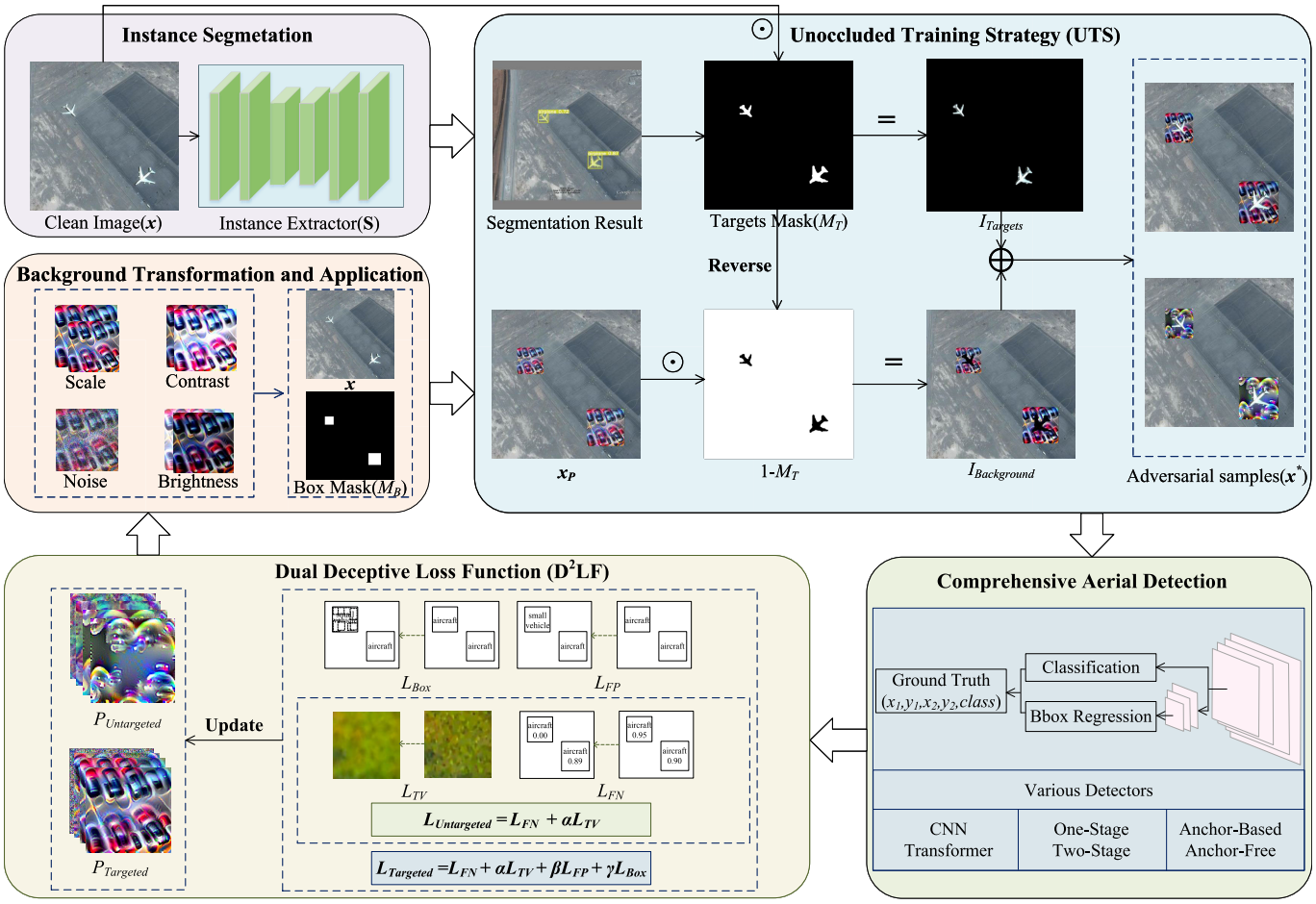


Fig. 2. Framework of the proposed BADEI. First, instance segmentation is employed to extract the target mask $\boldsymbol{M}_T$ and inverse-mask $(1 - \boldsymbol{M}_T)$ from the clean image $\boldsymbol{x}$. Second, BT is applied to enhance the robustness of the adversarial background, and the background application is appropriately used to place the background at the location of the target detection box, resulting in $\boldsymbol{x}_P$. Third, combining $\boldsymbol{M}_T$, $1 - \boldsymbol{M}_T$, and $\boldsymbol{x}_P$, an adversarial sample $\boldsymbol{x}^*$ is obtained with the background placed directly beneath the targets. Fourth, feed $\boldsymbol{x}^*$ into diverse aerial detectors for detection. Finally, update the pixel values of the background based on the proposed D^{2}LF specifically designed for dual-induced attacks.

targets aircraft but is not limited to them as subjects of interest. An overview of the entire process is illustrated in Fig. 2. Next, a detailed explanation of the UTS and D^{2}LF is provided.

A. Problem Formulation

In this work, the objective of adversarial attacks in aerial detection is to hide specified targets by placing elaborated adversarial backgrounds beneath the targets in clean images. Specifically, given a benign aerial image $\boldsymbol{x}$ and an instance segmentation extractor $\boldsymbol{S}$, the mask matrix of the targets $\boldsymbol{M}_T$ can be obtained through $\boldsymbol{S}$ and binarization (using the YOLOv5x-seg model for instance segmentation), where the pixel values of the foreground are 1, and the rest are 0. Next, initialize background patch $\boldsymbol{P}^0$ randomly and then apply the adversarial background $\boldsymbol{P}$ ($\boldsymbol{P}_{\text{untargeted}}$ or $\boldsymbol{P}_{\text{targeted}}$) to $\boldsymbol{x}$ by background application (BA) yielding $\boldsymbol{x}_P$, which can be denoted as follows:

$$\boldsymbol{x}_P = \text{BA}(\boldsymbol{x}, \boldsymbol{P}, l, w, h) \tag{1}$$

where (l, w, h) are the placement position of $\boldsymbol{P}$ and BA is the placement method. (l, w, h) is obtained by ground truth, and then, BA can further gain the box mask $(\boldsymbol{M}_B)$ for placing the adversarial background according to it. This will be elaborated on later.

Considering the above content, the adversarial example $\boldsymbol{x}^*$ with adversarial background $\boldsymbol{P}$ can be formulated as follows:

$$\boldsymbol{x}^* = \boldsymbol{x} \odot \boldsymbol{M}_T + \boldsymbol{x}_P \odot (\boldsymbol{1} - \boldsymbol{M}_T) \tag{2}$$

where $\odot$ represents the Hadamard product and $\boldsymbol{P}$ can be either the targeted background or the nontargeted background, respectively.

The existing approaches often aim to enhance the attack effectiveness by optimizing all pixels of the adversarial patch through strategies like placing it on or outside objects [21]. Recent research findings, such as CBA [22], have shown promising results in the physical world, but there still are challenges with the attack performance on a few of the detectors. Besides, the training strategy limits the improvement of patch's adversarial robustness in the digital domain. In comparison, our method fools aerial detectors by BADEI, which simultaneously elaborates FNs and FPs to conceal and misclassify the targets, as shown in Fig. 2. The UTS and D^{2}LF are developed to fool various aerial detectors and acquire a stronger attack efficacy both in the digital and physical world. In the following, we will systematically

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introduce how to perform dual-induced background attacks against aerial detection with aircraft as the protected objects by our proposed BADEI framework.

The workflow of the proposed BADEI method is outlined in Fig. 2. The goal of this approach is to generate adversarial backgrounds that are robust enough for real-world scenarios and the digital domain, preventing targets from being detected by aerial detectors, including CNN-based detectors, Transformer-based detectors, one-stage detector, two-stage detector, anchor-based detectors, and anchor-free detectors. In this work, we take aircraft as an example and place these adversarial backgrounds directly beneath the aircraft, thus avoiding smearing or obscuring the targets. To adapt to the scale variation of aircraft in remote sensing images and accommodate the dynamic conditions encountered in natural scenes, we adapt the adversarial background optimization process by incorporating various physical background transformations (BTs), including changes in lighting, scales, natural noises, and contrast. The formula is presented as follows:

$$\boldsymbol{P}_{i,j} = \mathrm{BT}\left(\boldsymbol{P}_{i,j}\right) \tag{3}$$

where $\boldsymbol{P}_{i,j}$ denotes the optimized adversarial background and i and j are the indices of epochs and iterations during training, respectively.

The adversarial background becomes adaptive and effective in different environmental conditions through (3). Then, a clean image $\boldsymbol{x}$ can obtain a mask matrix $\boldsymbol{M}_T$ via the instance extractor $\boldsymbol{S}$ and binarization, which is defined as follows:

$$\boldsymbol{M}_T = \boldsymbol{S}(\boldsymbol{x}) \tag{4}$$

where the original formulated problem (2) can be transformed as follows:

$$\boldsymbol{x}^* = \boldsymbol{x} \odot \boldsymbol{S}(\boldsymbol{x}) + \boldsymbol{x}_{\mathrm{BT}(\boldsymbol{P}_{i,j})} \odot (\boldsymbol{1} - \boldsymbol{S}(\boldsymbol{x})). \tag{5}$$

As shown in Fig. 2, according to the training strategy of adversarial background and loss function built on dual induction, the background pixels are used to iteratively update the pixel values of the adversarial background.

B. Unoccluded Training Strategy

Targets in remote sensing imagery are often small, and the training strategy of [21] is to place the patches on the targets, which easily leads to target occlusion and affects the camouflage effect. While in [22], the patches are placed outside the targets during the training process, in the test experiments, the patches are placed under the targets, which leads to the fact that the hiding effect still needs to be improved. It can be seen that the position of the adversarial background in the training strategy and experimental tests is crucial for the attack effectiveness. We introduce the UTS to place adversarial background directly below the targets for achieving adversarial background elaboration, as shown in Fig. 2, which optimizes and evaluates the adversarial backgrounds in the digital domain and performs physical attacks by printing the adversarial backgrounds and placing them physically in the scene of the attack.

Specifically, the elaborated adversarial background needs to be pasted on the target bounding box of the benign image with the proper size and location to obtain $\boldsymbol{x}_{\boldsymbol{P}}$. Because the in-class differences between aircraft and viewpoints often cause occlusion, the BADEI method applies adversarial backgrounds to clean images through BA, which serves as a foundation for later placing backgrounds beneath the targets without obstructing them. BA adjusts the coordination $\boldsymbol{l}_{\boldsymbol{P}}$ and size of the adversarial background $(w_{\boldsymbol{P}}, h_{\boldsymbol{P}})$ based on the ground truth $\boldsymbol{G} = (x_1, y_1, x_2, y_2, \mathrm{class})$ to place it at the center of the target bounding box. They can be represented by the following formulas:

$$\boldsymbol{l}_{\boldsymbol{P}_{i,j}} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \tag{6}$$

$$r_s = \frac{w_{\boldsymbol{P}_{i,j}} \cdot h_{\boldsymbol{P}_{i,j}}}{w_t \cdot h_t} \tag{7}$$

where w_t and h_t signify the size of the target detection box and r_s is the area ratio between the adversarial background and the object detection box.

Therefore, the size of $\boldsymbol{P}$ can be represented by the following formula:

$$w_{\boldsymbol{P}_{i,j}} = h_{\boldsymbol{P}_{i,j}} = \sqrt[2]{r_s \cdot w_t \cdot h_t}. \tag{8}$$

Furthermore, the corresponding box mask can be acquired based on $(\boldsymbol{l}_{\boldsymbol{P}_{i,j}}, w_{\boldsymbol{P}_{i,j}}, h_{\boldsymbol{P}_{i,j}})$ obtained above, and thus, the image $\boldsymbol{x}_{\boldsymbol{P}}$ with the background applied can be denoted as follows:

$$\boldsymbol{x}_{\boldsymbol{P}_{i,j}} = \mathrm{BA}\left(\boldsymbol{x}, \boldsymbol{P}_{i,j}, \boldsymbol{l}_{\boldsymbol{P}_{i,j}}, w_{\boldsymbol{P}_{i,j}}, h_{\boldsymbol{P}_{i,j}}\right). \tag{9}$$

Taking into account the mask matrix obtained from the instance extractor, the adversarial example $\boldsymbol{x}^*$ (5) is defined as follows:

$$\boldsymbol{x}^* = \boldsymbol{x} \odot \boldsymbol{S}(\boldsymbol{x}) + \mathrm{BA}\left(\boldsymbol{x}, \mathrm{BT}\left(\boldsymbol{P}_{i,j}\right), \quad \boldsymbol{l}_{\boldsymbol{P}_{i,j}}, w_{\boldsymbol{P}_{i,j}}, h_{\boldsymbol{P}_{i,j}}\right) \odot (\boldsymbol{1} - \boldsymbol{S}(\boldsymbol{x})). \tag{10}$$

The UTS we proposed exploits global contextual background information to attack aerial detectors and places adversarial backgrounds beneath the target objects, which is rigorously validated for its effectiveness through extensive experiments in Section IV.

C. Dual Deceptive Loss Function

The objective of this article is to achieve both nontargeted and targeted attacks through FNs and FPs via dual adversarial induction. To conceal the targets or classify them into a specified category, the D^{2}LF is designed to update the pixel values of the adversarial background. The objectiveness function consists of adversarial objectiveness loss, bounding box loss, and smoothness loss.

The adversarial objectiveness loss consists of two components: one is aimed at avoiding the detection of the aircraft (L_{FN}), and the other is focused on recognizing it as the specified category (L_{FP}). The D^{2}LF solves the optimization problem of the adversarial background that includes the minimum objectiveness score and the maximum specified category score.

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1) Adversarial Objectiveness L_{FN}: Let D be an aerial detector and $\boldsymbol{r}$ is the detection results, i.e., $\boldsymbol{r} = D(\boldsymbol{x}^*)$, where $\boldsymbol{r}$ usually includes coordinates (x_1, y_1, x_2, y_2), the objective score $P(\text{obj})$, and the class probability vector, such as $(P(\text{aircraft}), P(\text{ship}), \ldots, P(\text{bridge}), P(\text{harbor}))$. Technically, we aim to conceal the target objects (aircraft) against aerial detection, which entails reducing the confidence score and class probability of the detected aircraft to 0. Therefore, considering all the objectiveness scores, L_{FN} can be expressed by the following formula:

$$L_{\text{FN}} = E(\boldsymbol{r}) = \frac{1}{n}\sum_{i=1}^{n} P_i(\text{obj}) \tag{11}$$

where n means the number of detected objects and $E(\boldsymbol{r})$ represents the objectiveness loss L_{FN} extracted from $\boldsymbol{r}$. The adversarial objectiveness loss gifts the adversarial background with attack efficacy during the optimization. The adversarial background optimized with L_{FN} has a significant visual difference from the attacked targets, but can well cause the objects to evade detection by the aerial detectors.

2) Adversarial Objectiveness L_{FP}: BADEI we developed supports untargeted attacks and targeted specificity attacks to purposefully deceive victim models with object-vanishing and object-misidentifying effects. To elaborate targeted adversarial backgrounds, it is crucial to constrain higher scores for the specified category to misclassify aircraft, we define the loss L_{FP} as follows:

$$L_{\text{FP}} = 1 - M(\boldsymbol{r}) = 1 - \frac{1}{m}\sum_{j=1}^{m} P_j(\text{obj}) \tag{12}$$

where m means the number of specified objects and the meaning of $\boldsymbol{r}$ is the same as that in L_{FN}. $M(\boldsymbol{r})$ denotes the extracting objective score for the specified category targets from $\boldsymbol{r}$. L_{FP} gifts the adversarial background with targeted attack efficacy during the optimization.

3) Bounding Box Loss L_{Box}: During the adversarial background optimization process, we aim to induce misclassification of the unoccluded region within the adversarial background. Therefore, we apply a constraint to the bounding box coordinates (x_1', y_1', x_2', y_2'), which represent the coordinates of the specified category target detected by aerial detectors, to make them approach the ground truth (x_1, y_1, x_2, y_2), which can be written as follows:

$$\begin{aligned} L_{\text{Box}} &= B(\boldsymbol{G}, \boldsymbol{r}) \\ &= \frac{1}{m}\left(\left|\left(x_1' - x_1\right)\right| + \left|y_1' - y_1\right| + \left|x_2' - x_2\right| + \left|y_2' - y_2\right|\right) \end{aligned} \tag{13}$$

where B represents the calculation based on ground-truth coordinates and detection results.

4) Smoothness Loss L_{TV}: To ensure that the optimizer prefers to generate adversarial backgrounds with smooth color transitions and these adversarial backgrounds can achieve successful attack after cross-domain transformation, we adopt the total variation [47] as the smoothness loss L_{TV} of the generated adversarial backgrounds. The formula is as follows:

$$L_{\text{TV}} = \sum_{i,j}\sqrt{\left(V_{i,j} - V_{i+1,j}\right)^2 + \left(V_{i,j} - V_{i,j+1}\right)^2} \tag{14}$$

Algorithm 1 BADEI

Input: Clean aerial image $\boldsymbol{x}$, detector D, ground truth $\boldsymbol{G}$, instance extractor $\boldsymbol{S}$, initialized adversarial background randomly $\boldsymbol{P}^0$, the adversarial attack loss function L, the number of epochs N_{epc} and the number of iterations of each epoch N_{itr}, hyperparameter $\alpha, \beta, \gamma, \eta$.

Output: Adversarial background $\boldsymbol{P}$.

$\boldsymbol{P} = \boldsymbol{P}^0$;
Extract aircraft mask matrix $\boldsymbol{M}_T$ from $\boldsymbol{x}$, $\boldsymbol{M}_T = \boldsymbol{S}(\boldsymbol{x})$;
Aircraft's background mask matrix: $\mathbf{1} - \boldsymbol{M}_T$;
for $i = 0$ to N_{epc} **do**
  for $j = 0$ to N_{itr} **do**
    $\boldsymbol{P}_{i,j} = BT(\boldsymbol{P}_{i,j})$
    $\boldsymbol{x}_{\boldsymbol{P}_{i,j}} = BA(\boldsymbol{x}, \boldsymbol{P}_{i,j}, \boldsymbol{l}_{\boldsymbol{P}_{i,j}}, w_{\boldsymbol{P}_{i,j}}, h_{\boldsymbol{P}_{i,j}})$;
    $\boldsymbol{x}^* = \boldsymbol{x} \odot \boldsymbol{M}_T + \boldsymbol{x}_{\boldsymbol{P}_{i,j}} \odot (\mathbf{1} - \boldsymbol{M}_T)$;
    $\boldsymbol{r} = D(\boldsymbol{x}^*)$;
    $L_{\text{FN}} = E(\boldsymbol{r})$;
    $L_{\text{FP}} = 1 - M(\boldsymbol{r})$;
    $L_{\text{Box}} = B(\boldsymbol{G}, \boldsymbol{r})$;
    $L_{Untargeted} = L_{\text{FN}} + \alpha \cdot L_{\text{TV}}$;
    $L_{Targeted} = L_{\text{FN}} + \alpha \cdot L_{\text{TV}} + \beta \cdot L_{\text{FP}} + \gamma \cdot L_{\text{Box}}$;
    $L = L_{Untargeted}$ or $L_{Targeted}$;
    $\boldsymbol{P}_{i,j+1} = \boldsymbol{P}_{i,j} + \eta \cdot \nabla_{\boldsymbol{P}_{i,j}} L$;
  end for
end for
$\boldsymbol{P} = \boldsymbol{P}_{N_{epc}, N_{itr}}$;
return $\boldsymbol{P}$.

where i and j are the indices of the row and column of the adversarial background, respectively, and $V_{i,j}$ denotes the pixel value. L_{TV} overcomes the problem of the image acquisition devices having difficulty capturing the value gaps between adjacent pixels.

Finally, the untargeted objectiveness loss $L_{\text{Untargeted}}$ is as follows:

$$L_{\text{Untargeted}} = L_{\text{FN}} + \alpha \cdot L_{\text{TV}} \tag{15}$$

where α is a balance parameter, which is set to 1.5 based on prior experience.

The targeted objectiveness loss L_{Targeted} is defined as follows:

$$L_{\text{Targeted}} = L_{\text{FN}} + \alpha \cdot L_{\text{TV}} + \beta \cdot L_{\text{FP}} + \gamma \cdot L_{\text{Box}} \tag{16}$$

where α, β, and γ are the hyperparameters that balance the loss terms. During the training process, α is set to 1.5, β is set to 1, and γ is set to 1. The selection of hyperparameters is discussed in detail in Section IV-D.

D. Algorithm

In this article, we select aircraft as the target objects to describe and validate the overall training process of the proposed BADEI, as illustrated in Algorithm 1. Specific details regarding this algorithm are provided as follows.

1) Extracting the aircraft mask matrix $\boldsymbol{M}_T$ from the instance extractor $\boldsymbol{S}$ and binarization to prevent occlusion of the aircraft, we place the adversarial background directly underneath the target.

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2) The adversarial backgrounds adapt to various environmental conditions through BT and then are adaptively placed at the target detection box location of the original image $\boldsymbol{x}$ by BA.
3) Combining the aircraft mask matrix $\boldsymbol{M}_T$ and the image $\boldsymbol{x}_P$ placed with adversarial backgrounds, we formulate the malicious instance $\boldsymbol{x}^*$, aiming to obtain images where backgrounds are placed beneath aircraft without occluding them.
4) Adversarial examples $\boldsymbol{x}^*$ are fed into the victim models D for detection, yielding detection results $\boldsymbol{r}$.
5) The objectiveness losses L_{FN} and L_{FP} and the bounding box loss L_{Box} are extracted from detection results $\boldsymbol{r}$ by the functions E, M, and B.
6) Integrated with the total variation loss L_{TV}, using L_{TV}, L_{FN} as the loss for nontargeted attack and L_{TV}, L_{FN}, L_{FP}, and L_{Box} as the loss for targeted attack, the pixel values of adversarial backgrounds $\boldsymbol{P}$ are updated based on specific loss function.
7) Proceed with steps 2–6 iteratively until the training process is concluded.

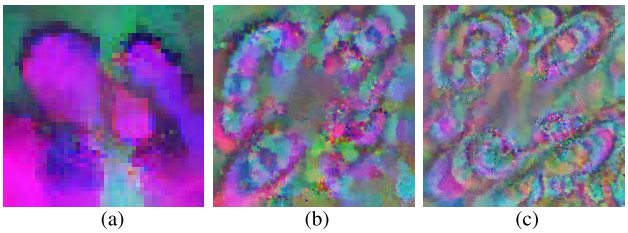
IV. Experiments

In this section, comprehensive and extensive experiments are conducted to demonstrate the effectiveness of our proposed method BADEI. First, the relevant experimental setup is presented. Subsequently, we provide a detailed exposition and analysis of the specific experiments and results from both the perspectives of nontargeted attack and targeted attack. Finally, we discuss the hyperparameter setting and ablation study.

A. Experimental Setup

1) Datasets: Dataset for object detection in aerial images (DOTA)[1] [55] and RSOD,[2] are used for training aerial detectors and our background perturbations, respectively. RSOD is a public dataset for aerial detection, which is downloaded, collected, and annotated by Wuhan University and contains four categories: aircraft, playground, overpass, and oil tank. DOTA is a large-scale dataset for object detection and object tracking tasks, which was created by Wuhan University and contains 15 categories: aircraft, baseball-diamond, bridge, ground-track-field, small vehicles, large vehicles, ship, tennis-court, basketball-court, storage-tank, soccer-ball-field, roundabout, harbor, swimming-pool, and helicopter. In this article, DOTA dataset, celebrated for its data richness, is utilized for training aerial detectors, while RSOD dataset is employed for the extraction of target masks and the optimization of adversarial backgrounds.

2) Victim Models: Based on the DOTA dataset, we benchmark the effectiveness of adversarial backgrounds under both white-box and black-box settings with various mainstream object detectors, including Single Shot MultiBox Detector (SSD) [56], Faster R-CNN [57], Swin Transformer [58], YOLOv3 [59], YOLOv5 [60], Cascade R-CNN [61], RetinaNet [62], Mask R-CNN [63], FreeAnchor [64], Feature Selective Anchor-Free (FSAF) [65], RepPoints [66], TOOD [67], Adaptive Training Sample Selection (ATSS) [68], FoveaBox [69], and VarifocalNet [70].



(a) (b) (c)

Fig. 3. Adversarial backgrounds at different resolutions. (a) 50. (b) 150. (c) 300.

3) Evaluation Metrics: We evaluate the effectiveness of the adversarial backgrounds through average precision (AP), which is defined as the AP of a category and calculated as follows:

$$\text{AP} = \int_0^1 p_i(r)dr \tag{17}$$

where $p_i(r)$ is the interpolated precision at a certain recall level r. It is defined as follows:

$$p_i(r) = \max_{\tilde{r} \geq r} \tilde{p}\left(\tilde{r}\right) \tag{18}$$

where $\tilde{p}(\tilde{r})$ is the precision at $\tilde{r}$ recall level. Such interpolation is conducted by taking the maximum precision value overall recall levels greater than or equal to r. The AP is calculated by averaging the precision values at all the recall levels at which there is a correct detection. Besides, we assess whether the FPs and FNs elaborated successfully conceal aircraft and identify specified target category by confidence scores.

4) Compared Methods: We select several state-of-the-art adversarial attack methods against aerial detection for comparison, including adversarial patches presented by Thys et al. [40], AP-PA [21], and CBA [22], to validate the effectiveness of the method we proposed.

5) Resolution Setting: We delve into the impact of the resolution of adversarial backgrounds on attack effectiveness. YOLOv5n is employed as a proxy model to train adversarial backgrounds for hiding targets. Fig. 3 shows different adversarial backgrounds with resolutions of 50, 150, and 300, respectively. The AP values detected by different aerial detectors are recorded and presented in Table I. From the table, it is evident that the effectiveness of our proposed method in attacking aerial detectors is to some extent influenced by the resolution of the adversarial backgrounds. In most cases, higher resolution adversarial backgrounds better evade detection. When attacking YOLOv5n and Swin Transformer, lower resolution adversarial backgrounds demonstrate marginally better performance, though the improvement is not considerable compared with higher resolution backgrounds. Therefore, in the experiments of this article, a resolution of 300 × 300 is adopted for the adversarial backgrounds.

6) Implementation Details: The adversarial background is randomly initialized and set to a size of 300 × 300 pixels. During training, Adam is adopted as the optimizer with an initial learning rate of 0.03, and the thresholds for the intersection of union (IOU) and objective confidence are set as 0.45 and 0.4. Besides, based on the prior work [22] and hyperparameter analysis, we set α, β, and γ to 0.03,

[1]https://captain-whu.github.io/DOTA/index.html
[2]https://github.com/RSIA-LIESMARS-WHU/RSOD-Dataset-

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TABLE I
COMPARISON RESULTS OF ADVERSARIAL BACKGROUNDS WITH DIFFERENT RESOLUTIONS

Resolution \ Detectors	YOLOv3	YOLOv5n	YOLOv5s	YOLOv5m	YOLOv5l	YOLOv5x	Faster R-CNN	SSD	Swin Transformer
50	0.484	**0.120**	0.453	0.512	0.498	0.508	0.127	**0.015**	**0.033**
150	0.490	0.253	0.488	0.496	0.517	0.518	0.085	0.038	0.043
300	**0.461**	0.165	**0.421**	**0.457**	**0.481**	**0.486**	**0.045**	**0.015**	0.053

YOLOv5 serves as the proxy model for generating adversarial backgrounds of various resolutions, with **bold** formatting indicating the best outcomes.

1, and 1, respectively. All digital experiments are performed using pytorch on NVIDIA GeForce RTX4090 24GB GPUs. Besides, experiments in the physical world are carried out at a scale of 1:400. It is worth noting that all experiments in the physical domain are conducted by using model aircraft that are proportionate to real aircraft.

B. Untargeted Attack

1) Elaborated Untargeted Adversarial Backgrounds: We train the adversarial backgrounds by directly pasting them beneath the targets (aircraft) detected by the aerial detector. In this process, we initially generate a random background and employ $L_{\text{Untargeted}}$ to generate FNs against the detection networks mentioned before. The elaborated adversarial backgrounds are shown in Fig. 4.

By comparing the adversarial backgrounds in Fig. 4, some intriguing properties can be observed. First, it can be observed that models with similar structures tend to produce adversarial backgrounds with more similar styles. For instance, the five models in the YOLOv5 series generate FNs that exhibit stylistic similarities. Similarly, FNs generated by two-stage detectors (Faster R-CNN and Cascade R-CNN) also exhibit resemblances in style. Conversely, detectors with significant structural differences yield adversarial backgrounds with more pronounced distinctions in both style and color, as seen in the cases of TOOD and VarifocalNet. Furthermore, we can note that the pattern variations on the adversarial backgrounds are primarily concentrated on the periphery of the backgrounds, with the central region showing minimal changes. This phenomenon is closely related to our training strategy, which avoids blocking the targets and then improves the attack effectiveness of practical testing.

2) Untargeted Attack in Digital World: The adversarial patches are affixed outside the objects detected by aerial detectors in proper size and positions in the previous work [22], limiting attack effectiveness in the digital domain. Based on this, a more practical training strategy is proposed. To better demonstrate the impact of FNs on aerial detectors, ten aerial detectors are still employed for quantitative evaluation, as shown in Table II, displaying the results of digital attacks.

By comparing the method introduced with others, it demonstrates significant advantages of our approach.

1) The adversarial backgrounds FNs meticulously crafted are employed to conduct adversarial attacks on ten aerial detectors. The results reveal that, in terms of aircraft detection, FN-based background attack exhibits significantly lower AP values compared to existing methods. In the table, "Ours(untargeted)" is highlighted in distinct red, indicating that the approach notably enhances the effectiveness of hidden targets in the digital domain.
2) Faster R-CNN, SSD, and Swin Transformer models are more susceptible to FN-based background attack, with their AP for aircraft detection approaching zero. In contrast, YOLOv3 and YOLOv5 exhibit higher adversarial robustness. Furthermore, as the convolutional layers in YOLOv5 increase, the model's representation capacity and robustness gradually improve, making it more challenging to succeed in nontargeted attacks. In general, FN-based background attack demonstrates superior attack effectiveness and transferability across these ten models.

3) Untargeted Attack in Physical Scenarios: During the physical experiments, pretrained adversarial backgrounds are printed and placed directly beneath each aircraft. Subsequently, the captured photographs are employed as inputs for 19 different aerial detectors. To ensure a precise quantitative analysis, a consistent threshold of 0.2 is maintained, and the average confidence scores for the detection results of 18 aircraft are computed. A comparison is conducted among several state-of-the-art methods, and the detailed comparison results are presented in Table III. As evident from the table, the following holds.

1) White-box attacks achieve a significant advantage in terms of attack effectiveness. Out of the adversarial backgrounds trained with the 19 detectors, 17 of them successfully resulted in the complete invisibility of the targets. In addition, the other two detect the targets with an average confidence score of less than 5%.
2) FN-based background attack demonstrates outstanding transferability in black-box attacks. Particularly, the adversarial background trained by VarifocalNet has been successful in rendering 18 different aerial detectors completely incapable of detecting the targets.
3) The previous methods have some limitations, particularly in terms of the limited effectiveness of adversarial backgrounds against YOLOv3 and YOLOv5 detectors. However, FNs have notably improved the attack efficacy against these two models in the physical world, which can be readily observed from the prominent red markings in the corresponding rows of the table.

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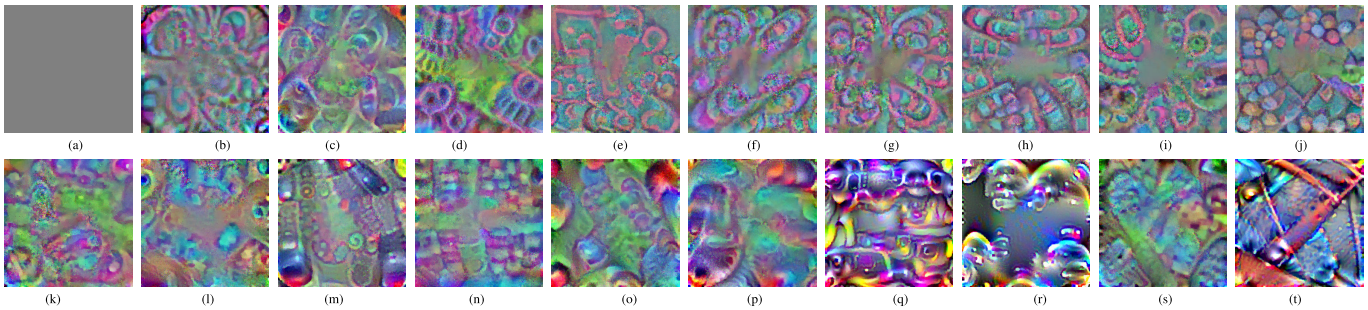


Fig. 4. Untargeted adversarial backgrounds FNs elaborated via background attack. (a) Random. (b) SSD. (c) Faster R-CNN. (d) Swin Transformer. (e) YOLOv3. (f) YOLOv5n. (g) YOLOv5s. (h) YOLOv5m. (i) YOLOv5l. (j) YOLOv5x. (k) Cascade R-CNN. (l) RetinaNet. (m) Mask R-CNN. (n) FreeAnchor. (o) FSAF. (p) RepPoints. (q) TOOD. (r) ATSS. (s) FoveaBox. (t) VarifocalNet.

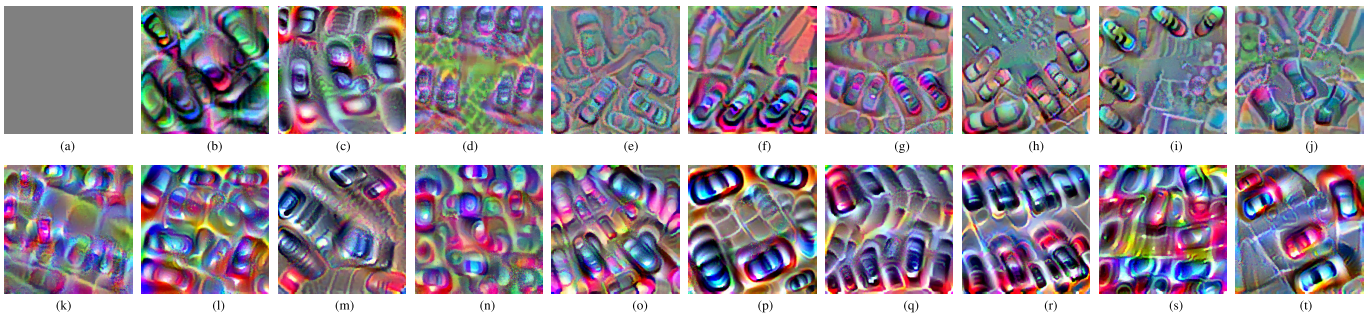


Fig. 5. Targeted adversarial backgrounds FPs elaborated via background attack. (a) Random. (b) SSD. (c) Faster R-CNN. (d) Swin Transformer. (e) YOLOv3. (f) YOLOv5n. (g) YOLOv5s. (h) YOLOv5m. (i) YOLOv5l. (j) YOLOv5x. (k) Cascade R-CNN. (l) RetinaNet. (m) Mask R-CNN. (n) FreeAnchor. (o) FSAF. (p) RepPoints. (q) TOOD. (r) ATSS. (s) FoveaBox. (t) VarifocalNet.

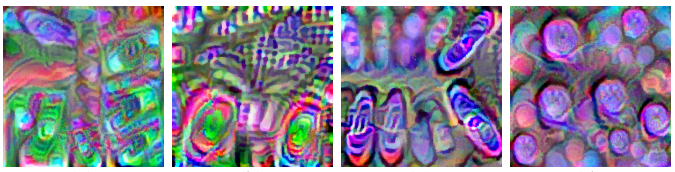


Fig. 6. Targeted adversarial backgrounds with various specified categories. (a)–(d) FPs that identify aircraft as basketball-courts, ground-track-fields, ships, and storage-tanks, respectively. (a) FP1. (b) FP2. (c) FP3. (d) FP4.

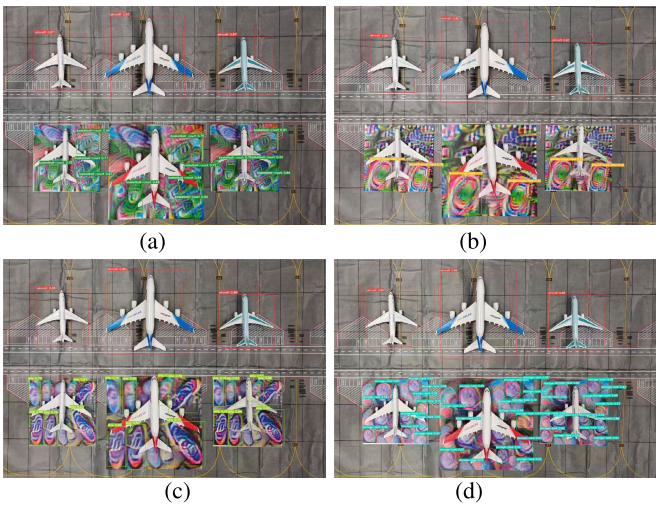


Fig. 7. Detection results with different FPs. (a) Basketball-court. (b) Ground-track-field. (c) Ship. (d) Storage-tank.

C. Targeted Attack

1) Elaborated Targeted Adversarial Backgrounds: Employing L_{Targeted} to generate adversarial backgrounds FPs for 19 aerial detectors, respectively, comprehensive experiments are conducted to demonstrate the effectiveness of hiding aircraft and recognizing them as small vehicles. The specific generated backgrounds are shown in Fig. 5.

By comparing the generated adversarial backgrounds, we can observe the similarities between FPs and FNs. Backgrounds generated by models with similar structures exhibit similar styles, such as YOLOv3 and YOLOv5. In addition, it is noticeable that the shape of adversarial backgrounds is closely related to the target category we aim to recognize.

Certainly, the proposed approach can also generate false positive (FP) adversarial backgrounds that can identify targets as other categories. As shown in Fig. 6, we present some adversarial backgrounds that can be recognized as other categories, reaffirming the correlation between the shape of targeted adversarial backgrounds and the specified categories. Simultaneously, we conduct experiments physically to validate the attack effectiveness of these adversarial backgrounds. The results are depicted in Fig. 7. It is obvious that aircraft placed with different adversarial backgrounds can be successfully hidden and identified as different categories of targets, but aircraft without adversarial backgrounds are still detected with high confidence scores.

2) Targeted Attack in Digital World: Similarly, in the digital domain, targeted attacks against aerial detectors are executed by affixing adversarial backgrounds underneath

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TABLE II
DIGITAL ATTACK RESULTS COMPARISON

		YOLOv3	YOLOv5n	YOLOv5s	YOLOv5m	YOLOv5l	YOLOv5x	Faster R-CNN	SSD	Swin Transformer
Thys et al.	YOLOv3	**0.631**	0.803	0.715	0.796	0.768	0.784	0.644	0.461	0.863
	YOLOv5n	0.704	**0.772**	0.668	0.810	0.768	0.790	0.266	0.338	0.855
	YOLOv5s	0.667	0.798	**0.671**	0.760	0.761	0.775	0.608	0.455	0.843
	YOLOv5m	0.696	0.806	0.708	**0.756**	0.773	0.802	0.620	0.472	0.848
	YOLOv5l	0.682	0.810	0.701	0.766	**0.753**	0.787	0.645	0.492	0.862
	YOLOv5x	0.695	0.819	0.697	0.775	0.768	**0.769**	0.649	0.445	0.838
	Faster R-CNN	0.681	0.776	0.739	0.777	0.802	0.819	**0.469**	0.387	0.848
	SSD	0.668	0.774	0.670	0.753	0.743	0.764	0.346	**0.248**	0.793
	Swin Transformer	0.926	0.899	0.906	0.935	0.937	0.952	0.568	0.608	**0.820**
APPA (on)	YOLOv3	**0.592**	0.754	0.664	0.742	0.725	0.754	0.354	0.289	0.828
	YOLOv5n	0.886	**0.839**	0.853	0.916	0.895	0.924	0.372	0.400	0.812
	YOLOv5s	0.652	0.783	**0.636**	0.751	0.741	0.765	0.537	0.381	0.838
	YOLOv5m	0.669	0.785	0.675	**0.735**	0.752	0.783	0.545	0.420	0.838
	YOLOv5l	0.655	0.783	0.672	0.742	**0.721**	0.753	0.564	0.412	0.843
	YOLOv5x	0.659	0.776	0.657	0.751	0.745	**0.739**	0.553	0.403	0.836
	Faster R-CNN	0.769	0.846	0.806	0.850	0.844	0.848	**0.296**	0.343	0.801
	SSD	0.725	0.815	0.778	0.802	0.806	0.817	0.310	**0.220**	0.790
	Swin Transformer	0.814	0.847	0.838	0.859	0.863	0.858	0.296	0.339	**0.707**
APPA (outside)	YOLOv3	**0.645**	0.570	0.669	0.674	0.728	0.681	0.464	0.494	0.579
	YOLOv5n	0.644	**0.394**	0.661	0.666	0.696	0.675	0.409	0.466	0.669
	YOLOv5s	0.687	0.581	**0.607**	0.704	0.736	0.715	0.346	0.480	0.594
	YOLOv5m	0.639	0.518	0.536	**0.552**	0.627	0.615	0.412	0.411	0.636
	YOLOv5l	0.671	0.580	0.574	0.639	**0.542**	0.639	0.340	0.426	0.631
	YOLOv5x	0.657	0.579	0.613	0.644	0.690	**0.627**	0.405	0.486	0.651
	Faster R-CNN	0.722	0.567	0.703	0.720	0.752	0.742	**0.303**	0.494	0.577
	SSD	0.685	0.541	0.660	0.629	0.713	0.671	0.426	**0.446**	0.629
	Swin Transformer	0.721	0.618	0.712	0.699	0.726	0.701	0.470	0.518	**0.579**
CBA	YOLOv3	**0.754**	0.730	0.769	0.763	0.782	0.776	0.380	0.547	0.663
	YOLOv5n	0.649	**0.508**	0.652	0.626	0.672	0.691	0.311	0.472	0.622
	YOLOv5s	0.718	0.631	**0.686**	0.686	0.743	0.743	0.341	0.494	0.667
	YOLOv5m	0.737	0.685	0.725	**0.683**	0.747	0.757	0.367	0.519	0.656
	YOLOv5l	0.726	0.651	0.715	0.703	**0.733**	0.747	0.389	0.508	0.668
	YOLOv5x	0.700	0.604	0.684	0.652	0.702	**0.682**	0.357	0.507	0.665
	Faster R-CNN	0.719	0.661	0.736	0.704	0.758	0.740	**0.288**	0.463	0.648
	SSD	0.692	0.628	0.709	0.702	0.751	0.741	0.341	**0.397**	0.610
	Swin Transformer	0.758	0.723	0.746	0.726	0.762	0.765	0.300	0.506	**0.604**
Ours(untargeted)	YOLOv3	**0.273**	0.231	0.397	0.430	0.462	0.464	0.050	0.012	0.052
	YOLOv5n	0.461	**0.165**	0.421	0.457	0.481	0.486	0.045	0.015	0.053
	YOLOv5s	0.409	0.114	**0.207**	0.372	0.411	0.446	0.033	0.012	0.033
	YOLOv5m	0.338	0.113	0.287	**0.204**	0.425	0.430	0.033	0.006	0.033
	YOLOv5l	0.329	0.136	0.293	0.308	**0.132**	0.435	0.029	0.011	0.043
	YOLOv5x	0.347	0.150	0.334	0.380	0.371	**0.274**	0.020	0.014	0.025
	Faster R-CNN	0.422	0.177	0.353	0.370	0.401	0.458	**0.000**	0.014	0.015
	SSD	0.436	0.230	0.384	0.413	0.460	0.468	0.027	**0.001**	0.043
	Swin Transformer	0.434	0.110	0.388	0.409	0.384	0.482	0.006	0.005	**0.001**
Ours(targeted)	YOLOv3	**0.218**	0.145	0.354	0.350	0.419	0.409	0.028	0.012	0.043
	YOLOv5n	0.379	**0.051**	0.341	0.417	0.458	0.448	0.021	0.013	0.022
	YOLOv5s	0.333	0.072	**0.116**	0.330	0.455	0.405	0.009	0.011	0.032
	YOLOv5m	0.287	0.210	0.323	**0.271**	0.431	0.409	0.026	0.015	0.031
	YOLOv5l	0.260	0.192	0.348	0.317	**0.275**	0.366	0.016	0.014	0.032
	YOLOv5x	0.320	0.164	0.338	0.380	0.384	**0.304**	0.006	0.012	0.025
	Faster R-CNN	0.396	0.163	0.355	0.408	0.433	0.432	**0.001**	0.012	0.014
	SSD	0.331	0.187	0.402	0.392	0.446	0.423	0.017	**0.001**	0.022
	Swin Transformer	0.413	0.048	0.275	0.360	0.413	0.426	0.004	0.011	**0.002**

1) The table presents FNs and FPs in the digital world under white-box and black-box attacks.
2) The light green area (first column) and light red area (second column) represent the adversarial attack methods and the aerial detectors employed for training adversarial backgrounds, while the light gray area (first row) denotes the aerial detectors used for testing.
3) For quantitative experiments, the detection result's AP value of clean images is considered as 100%. The color in the table is associated with the corresponding AP value. Value closer to 0 is indicated by red color, reflecting better attack performance of the adversarial background, while value closer to 1 is indicated by blue color, signifying poorer attack performance of the adversarial background.

aircraft. A record of the AP values is kept for the results of detecting aircraft, as depicted in Table II. It is observed that the following holds.

1) FPs still yield the most effective attacks on Faster R-CNN, SSD, and Swin Transformer among the ten aerial detectors. In addition, in both black-box and white-box attack scenarios, our adversarial backgrounds demonstrate outstanding attack capabilities.
2) "Ours(targeted)" appears prominently in red under target-specific attacks, effectively concealing the target from detection by aerial detectors. In some cases, FP-based background attack even outperforms FN-based background attack in their impact on YOLOv3, YOLOv5n, and YOLOv5s.

3) Targeted Attack in Physical World: During the targeted physical experiments, the adversarial backgrounds in Fig. 5 are employed and printed by the same printer. Subsequently, these backgrounds are positioned directly beneath the targets for object detection. The expected outcome of the detection is the incapacity of the aerial detectors to identify the targets, while potentially misclassifying the existing targets as the specified category. In the practical experiments involving targeted attacks, we conduct trials adopting aircraft as

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TABLE III

PHYSICAL UNTARGETED ATTACK RESULTS COMPARISON

		D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19
	Clean	0.996	0.996	0.998	0.897	0.816	0.901	0.904	0.911	0.909	0.999	0.988	0.933	0.988	0.851	0.875	0.819	0.710	0.923	0.921
Thys et al.	SSD (D1)	0.179	0.766	0.931	0.726	0.673	0.846	0.792	0.826	0.838	0.851	0.379	0.449	0.744	0.536	0.398	0.447	0.544	0.641	0.701
	Faster R-CNN (D2)	0.350	0.752	0.886	0.798	0.708	0.812	0.758	0.854	0.845	0.839	0.334	0.680	0.708	0.583	0.411	0.523	0.472	0.577	0.650
	Swin Transformer (D3)	0.582	0.756	0.982	0.840	0.741	0.793	0.701	0.820	0.820	0.846	0.586	0.853	0.888	0.592	0.581	0.511	0.514	0.799	0.776
	YOLOv3 (D4)	0.139	0.813	0.922	0.805	0.716	0.804	0.840	0.862	0.817	0.775	0.506	0.526	0.792	0.591	0.375	0.473	0.473	0.573	0.574
	YOLOv5n (D5)	0.519	0.893	0.967	0.764	0.746	0.804	0.861	0.856	0.848	0.927	0.818	0.876	0.914	0.653	0.554	0.534	0.546	0.812	0.794
	YOLOv5s (D6)	0.304	0.782	0.932	0.771	0.761	0.777	0.841	0.882	0.832	0.616	0.630	0.780	0.825	0.602	0.443	0.507	0.452	0.544	0.564
	YOLOv5m (D7)	0.377	0.813	0.938	0.817	0.706	0.826	0.834	0.820	0.811	0.729	0.659	0.852	0.822	0.703	0.510	0.594	0.534	0.576	0.677
	YOLOv5l (D8)	0.458	0.859	0.974	0.798	0.806	0.828	0.861	0.891	0.844	0.779	0.700	0.856	0.859	0.666	0.490	0.591	0.539	0.586	0.669
	YOLOv5x (D9)	0.367	0.877	0.942	0.736	0.674	0.815	0.849	0.852	0.823	0.630	0.661	0.871	0.831	0.669	0.468	0.544	0.493	0.513	0.627
	Cascade R-CNN (D10)	0.544	0.629	0.967	0.823	0.646	0.792	0.820	0.829	0.847	0.873	0.642	0.393	0.817	0.455	0.259	0.277	0.425	0.515	0.574
	RetinaNet (D11)	0.606	0.918	0.948	0.827	0.772	0.826	0.832	0.871	0.850	0.823	0.785	0.892	0.928	0.694	0.552	0.552	0.542	0.725	0.726
	Mask R-CNN (D12)	0.787	0.921	0.966	0.873	0.790	0.835	0.840	0.893	0.894	0.929	0.891	0.867	0.964	0.717	0.634	0.603	0.565	0.809	0.812
	FreeAnchor (D13)	0.351	0.749	0.942	0.814	0.729	0.793	0.833	0.889	0.873	0.844	0.616	0.543	0.827	0.534	0.361	0.408	0.503	0.528	0.599
	FSAF (D14)	0.399	0.710	0.944	0.802	0.756	0.827	0.817	0.847	0.892	0.906	0.699	0.477	0.889	0.528	0.402	0.408	0.487	0.792	0.744
	RepPoints (D15)	0.861	0.901	0.943	0.823	0.769	0.828	0.833	0.849	0.853	0.894	0.789	0.671	0.918	0.613	0.410	0.403	0.499	0.760	0.749
	TOOD (D16)	0.384	0.463	0.959	0.811	0.728	0.811	0.734	0.821	0.871	0.860	0.687	0.251	0.786	0.479	0.246	0.214	0.372	0.476	0.532
	ATSS (D17)	0.554	0.890	0.944	0.821	0.693	0.812	0.829	0.841	0.855	0.914	0.751	0.628	0.892	0.610	0.411	0.413	0.505	0.758	0.789
	FoveaBox (D18)	0.500	0.717	0.919	0.802	0.669	0.786	0.783	0.790	0.832	0.747	0.696	0.545	0.840	0.558	0.510	0.484	0.510	0.664	0.635
	VarifocalNet (D19)	0.392	0.622	0.943	0.792	0.692	0.813	0.819	0.804	0.851	0.758	0.518	0.411	0.784	0.498	0.247	0.347	0.494	0.536	0.497
APPA (on)	SSD (D1)	0.246	0.830	0.967	0.838	0.703	0.853	0.780	0.856	0.836	0.896	0.595	0.574	0.850	0.606	0.451	0.500	0.567	0.747	0.708
	Faster R-CNN (D2)	0.473	0.503	0.990	0.806	0.764	0.852	0.754	0.856	0.854	0.912	0.364	0.396	0.847	0.442	0.248	0.252	0.423	0.498	0.571
	Swin Transformer (D3)	0.204	0.682	0.867	0.676	0.652	0.750	0.663	0.778	0.784	0.853	0.518	0.663	0.680	0.479	0.319	0.378	0.389	0.580	0.556
	YOLOv3 (D4)	0.632	0.903	0.993	0.851	0.803	0.874	0.823	0.878	0.872	0.832	0.882	0.783	0.873	0.678	0.577	0.551	0.569	0.751	0.736
	YOLOv5n (D5)	0.754	0.609	0.993	0.813	0.704	0.836	0.733	0.830	0.852	0.884	0.804	0.603	0.921	0.481	0.379	0.358	0.451	0.669	0.689
	YOLOv5s (D6)	0.323	0.908	0.943	0.763	0.755	0.821	0.864	0.859	0.862	0.823	0.604	0.768	0.786	0.586	0.382	0.487	0.523	0.523	0.614
	YOLOv5m (D7)	0.423	0.829	0.926	0.776	0.786	0.778	0.863	0.849	0.854	0.748	0.652	0.802	0.847	0.632	0.461	0.521	0.515	0.563	0.701
	YOLOv5l (D8)	0.398	0.849	0.987	0.807	0.756	0.857	0.878	0.874	0.838	0.838	0.650	0.889	0.859	0.712	0.494	0.581	0.572	0.626	0.684
	YOLOv5x (D9)	0.307	0.842	0.986	0.812	0.765	0.825	0.857	0.882	0.877	0.855	0.703	0.825	0.842	0.662	0.492	0.603	0.555	0.647	0.638
	Cascade R-CNN (D10)	0.535	0.930	0.915	0.832	0.734	0.806	0.824	0.830	0.866	0.911	0.810	0.813	0.912	0.646	0.543	0.526	0.529	0.779	0.731
	RetinaNet (D11)	0.407	0.810	0.942	0.811	0.733	0.786	0.812	0.829	0.831	0.601	0.603	0.790	0.776	0.622	0.307	0.481	0.474	0.415	0.588
	Mask R-CNN (D12)	0.730	0.931	0.954	0.857	0.784	0.823	0.840	0.891	0.896	0.893	0.892	0.839	0.945	0.742	0.589	0.568	0.573	0.779	0.781
	FreeAnchor (D13)	0.614	0.926	0.944	0.857	0.815	0.804	0.840	0.856	0.843	0.806	0.870	0.873	0.894	0.682	0.554	0.537	0.542	0.641	0.692
	FSAF (D14)	0.494	0.768	0.941	0.809	0.742	0.838	0.824	0.836	0.844	0.831	0.769	0.463	0.821	0.588	0.394	0.457	0.513	0.768	0.747
	RepPoints (D15)	0.438	0.762	0.941	0.798	0.738	0.789	0.784	0.864	0.871	0.835	0.612	0.539	0.869	0.577	0.346	0.385	0.467	0.523	0.567
	TOOD (D16)	0.423	0.509	0.932	0.814	0.707	0.779	0.819	0.802	0.860	0.892	0.729	0.260	0.835	0.406	0.247	0.179	0.437	0.514	0.531
	ATSS (D17)	0.621	0.713	0.943	0.796	0.720	0.786	0.823	0.846	0.855	0.906	0.703	0.468	0.874	0.544	0.381	0.348	0.449	0.587	0.584
	FoveaBox (D18)	0.495	0.861	0.976	0.823	0.764	0.828	0.833	0.823	0.844	0.941	0.845	0.534	0.925	0.667	0.519	0.383	0.532	0.774	0.750
	VarifocalNet (D19)	0.725	0.876	0.937	0.856	0.807	0.833	0.837	0.833	0.864	0.870	0.798	0.877	0.903	0.627	0.614	0.541	0.516	0.707	0.701
APPA (outside)	SSD (D1)	0.929	0.941	0.978	0.819	0.772	0.820	0.834	0.875	0.896	0.907	0.883	0.739	0.929	0.762	0.683	0.619	0.622	0.813	0.837
	Faster R-CNN (D2)	0.947	0.968	0.928	0.870	0.788	0.874	0.894	0.893	0.901	0.927	0.934	0.603	0.979	0.749	0.671	0.667	0.616	0.828	0.858
	Swin Transformer (D3)	0.886	0.938	0.889	0.868	0.797	0.849	0.836	0.868	0.892	0.907	0.857	0.763	0.929	0.733	0.701	0.657	0.588	0.778	0.829
	YOLOv3 (D4)	0.836	0.968	0.949	0.774	0.733	0.816	0.801	0.845	0.879	0.924	0.883	0.661	0.929	0.734	0.579	0.636	0.599	0.824	0.817
	YOLOv5n (D5)	0.917	0.993	0.975	0.869	0.764	0.844	0.860	0.885	0.901	0.960	0.963	0.855	0.989	0.815	0.791	0.720	0.647	0.881	0.882
	YOLOv5s (D6)	0.868	0.934	0.938	0.819	0.806	0.834	0.861	0.885	0.894	0.912	0.878	0.779	0.910	0.777	0.699	0.676	0.636	0.841	0.859
	YOLOv5m (D7)	0.934	0.991	0.951	0.867	0.797	0.847	0.872	0.886	0.899	0.952	0.879	0.849	0.936	0.787	0.837	0.753	0.647	0.819	0.853
	YOLOv5l (D8)	0.951	0.987	0.940	0.881	0.804	0.884	0.877	0.892	0.903	0.992	0.951	0.927	0.983	0.803	0.758	0.724	0.642	0.851	0.898
	YOLOv5x (D9)	0.932	0.987	0.943	0.871	0.774	0.837	0.875	0.884	0.896	0.935	0.920	0.854	0.979	0.801	0.783	0.719	0.629	0.834	0.876
	Cascade R-CNN (D10)	0.985	0.942	0.939	0.862	0.802	0.836	0.841	0.893	0.880	0.934	0.905	0.800	0.943	0.780	0.746	0.682	0.614	0.793	0.847
	RetinaNet (D11)	0.982	0.960	0.937	0.787	0.789	0.824	0.831	0.837	0.837	0.931	0.899	0.827	0.943	0.763	0.767	0.691	0.607	0.780	0.826
	Mask R-CNN (D12)	0.963	0.942	0.944	0.832	0.795	0.842	0.843	0.843	0.848	0.926	0.860	0.743	0.942	0.780	0.707	0.670	0.627	0.832	0.843
	FreeAnchor (D13)	0.973	0.938	0.943	0.872	0.771	0.842	0.846	0.877	0.896	0.930	0.909	0.855	0.941	0.764	0.759	0.703	0.633	0.814	0.846
	FSAF (D14)	0.982	0.940	0.942	0.839	0.784	0.842	0.845	0.846	0.888	0.933	0.931	0.776	0.943	0.723	0.739	0.636	0.613	0.810	0.842
	RepPoints (D15)	0.971	0.942	0.943	0.831	0.805	0.839	0.846	0.844	0.894	0.939	0.957	0.847	0.944	0.784	0.680	0.676	0.649	0.823	0.863
	TOOD (D16)	0.978	0.942	0.944	0.882	0.801	0.845	0.866	0.871	0.881	0.938	0.941	0.867	0.942	0.807	0.769	0.704	0.666	0.822	0.864
	ATSS (D17)	0.951	0.936	0.955	0.839	0.778	0.834	0.846	0.826	0.843	0.933	0.872	0.821	0.917	0.762	0.722	0.691	0.603	0.808	0.820
	FoveaBox (D18)	0.972	0.953	0.944	0.847	0.794	0.834	0.865	0.862	0.843	0.940	0.939	0.814	0.943	0.763	0.704	0.701	0.608	0.781	0.829
	VarifocalNet (D19)	0.984	0.940	0.943	0.839	0.791	0.836	0.842	0.870	0.873	0.906	0.904	0.795	0.943	0.757	0.717	0.691	0.642	0.829	0.847
CBA	SSD (D1)	0.000	0.000	0.306	0.057	0.000	0.271	0.343	0.251	0.294	0.052	0.000	0.000	0.000	0.000	0.000	0.012	0.058	0.014	0.039
	Faster R-CNN (D2)	0.037	0.264	0.129	0.434	0.228	0.444	0.518	0.461	0.736	0.215	0.065	0.166	0.276	0.057	0.051	0.140	0.167	0.119	0.383
	Swin Transformer (D3)	0.000	0.341	0.022	0.443	0.104	0.472	0.449	0.526	0.716	0.282	0.069	0.217	0.151	0.057	0.036	0.248	0.260	0.199	0.343
	YOLOv3 (D4)	0.000	0.063	0.062	0.014	0.194	0.502	0.643	0.359	0.536	0.318	0.000	0.031	0.092	0.000	0.000	0.128	0.248	0.000	0.246
	YOLOv5n (D5)	0.082	0.128	0.029	0.202	0.000	0.206	0.391	0.198	0.198	0.000	0.000	0.208	0.034	0.000	0.000	0.082	0.147	0.000	0.000
	YOLOv5s (D6)	0.041	0.472	0.382	0.110	0.034	0.102	0.489	0.159	0.116	0.000	0.000	0.456	0.000	0.174	0.000	0.236	0.281	0.064	0.000
	YOLOv5m (D7)	0.028	0.336	0.147	0.373	0.020	0.312	0.422	0.218	0.317	0.035	0.000	0.282	0.079	0.055	0.000	0.199	0.249	0.072	0.016
	YOLOv5l (D8)	0.022	0.347	0.097	0.056	0.199	0.314	0.418	0.013	0.218	0.299	0.027	0.304	0.044	0.103	0.000	0.201	0.219	0.049	0.102
	YOLOv5x (D9)	0.000	0.409	0.047	0.131	0.068	0.187	0.413	0.112	0.057	0.000	0.000	0.267	0.031	0.000	0.000	0.124	0.159	0.012	0.000
	Cascade R-CNN (D10)	0.000	0.000	0.191	0.422	0.000	0.539	0.601	0.483	0.549	0.000	0.000	0.000	0.000	0.000	0.000	0.026	0.101	0.000	0.117
	RetinaNet (D11)	0.057	0.312	0.238	0.429	0.353	0.721	0.699	0.694	0.760	0.169	0.000	0.262	0.041	0.184	0.026	0.326	0.274	0.000	0.347
	Mask R-CNN (D12)	0.268	0.072	0.232	0.645	0.436	0.499	0.358	0.616	0.646	0.117	0.516	0.126	0.548	0.057	0.000	0.082	0.278	0.310	0.656
	FreeAnchor (D13)	0.051	0.151	0.215	0.408	0.000	0.410	0.389	0.389	0.517	0.129	0.000	0.147	0.000	0.016	0.000	0.078	0.146	0.000	0.180
	FSAF (D14)	0.016	0.112	0.038	0.732	0.332	0.787	0.726	0.690	0.802	0.000	0.000	0.120	0.013	0.000	0.000	0.133	0.186	0.018	0.202
	RepPoints (D15)	0.016	0.157	0.081	0.409	0.215	0.364	0.364	0.449	0.429	0.152	0.017	0.078	0.026	0.060	0.000	0.039	0.144	0.077	0.206
	TOOD (D16)	0.081	0.071	0.162	0.214	0.278	0.522	0.410	0.439	0.603	0.011	0.053	0.000	0.144	0.000	0.000	0.000	0.147	0.000	0.423
	ATSS (D17)	0.000	0.139	0.139	0.471	0.258	0.567	0.648	0.466	0.548	0.016	0.000	0.093	0.061	0.026	0.000	0.134	0.074	0.000	0.053
	FoveaBox (D18)	0.000	0.197	0.093	0.806	0.215	0.753	0.664	0.782	0.842	0.012	0.000	0.412	0.080	0.038	0.000	0.279	0.274	0.000	0.255
	VarifocalNet (D19)	0.210	0.382	0.071	0.618	0.402	0.642	0.663	0.644	0.574	0.062	0.031	0.129	0.035	0.042	0.029	0.332	0.274	0.034	0.438
Ours(untargeted)	SSD (D1)	0.034	0.216	0.258	0.262	0.040	0.085	0.228	0.074	0.246	0.310	0.101	0.169	0.231	0.097	0.000	0.084	0.158	0.326	0.297
	Faster R-CNN (D2)	0.040	0.000	0.142	0.037	0.000	0.046	0.154	0.042	0.046	0.048	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.023
	Swin Transformer (D3)	0.033	0.028	0.000	0.065	0.000	0.065	0.037	0.037	0.033	0.000	0.000	0.000	0.034	0.000	0.000	0.000	0.026	0.000	0.158
	YOLOv3 (D4)	0.000	0.129	0.093	0.000	0.071	0.055	0.028	0.161	0.143	0.456	0.073	0.173	0.261	0.023	0.000	0.078	0.222	0.118	0.387
	YOLOv5n (D5)	0.049	0.144	0.084	0.030	0.023	0.083	0.154	0.047	0.189	0.302	0.146	0.101	0.199	0.086	0.000	0.148	0.197	0.232	0.235
	YOLOv5s (D6)	0.023	0.044	0.179	0.000	0.044	0.000	0.023	0.000	0.000	0.183	0.146	0.000	0.117	0.000	0.000	0.000	0.056	0.094	0.139
	YOLOv5m (D7)	0.048	0.173	0.208	0.000	0.000	0.000	0.000	0.000	0.034	0.214	0.153	0.134	0.274	0.050	0.025	0.115	0.217	0.131	0.393
	YOLOv5l (D8)	0.083	0.215	0.055	0.024	0.046	0.075	0.039	0.000	0.092	0.319	0.216	0.081	0.258	0.034	0.000	0.111	0.274	0.167	0.389
	YOLOv5x (D9)	0.069	0.097	0.263	0.044	0.037	0.036	0.044	0.044	0.000	0.324	0.038	0.072	0.000	0.024	0.000	0.000	0.111	0.108	0.440
	Cascade R-CNN (D10)	0.092	0.000	0.000	0.129	0.000	0.148	0.267	0.159	0.192	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.052	0.000	0.097
	RetinaNet (D11)	0.096	0.142	0.259	0.446	0.159	0.124	0.752	0.621	0.738	0.100	0.000	0.206	0.031	0.032	0.000	0.071	0.184	0.000	0.272
	Mask R-CNN (D12)	0.151	0.000	0.281	0.123	0.000	0.044	0.124	0.244	0.158	0.055	0.115	0.000	0.116	0.000	0.000	0.000	0.000	0.109	0.057
	FreeAnchor (D13)	0.105	0.000	0.264	0.221	0.066	0.042	0.361	0.283	0.264	0.094	0.000	0.000	0.000	0.000	0.000	0.000	0.026	0.045	0.084
	FSAF (D14)	0.192	0.023	0.126	0.329	0.031	0.087	0.465	0.589	0.763	0.055	0.069	0.037	0.135	0.000	0.000	0.026	0.054	0.049	0.071
	RepPoints (D15)	0.029	0.000	0.101	0.038	0.000	0.049	0.156	0.038	0.048	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	TOOD (D16)	0.000	0.000	0.000	0.000	0.026	0.000	0.036	0.031	0.000	0.053	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.034
	ATSS (D17)	0.048	0.000	0.102	0.000	0.033	0.028	0.148	0.000	0.030	0.000	0.000	0.000	0.032	0.000	0.000	0.000	0.000	0.000	0.059
	FoveaBox (D18)	0.041	0.052	0.025	0.000	0.000	0.034	0.248	0.140	0.081	0.044	0.026	0.000	0.030	0.000	0.000	0.047	0.066	0.000	0.070
	VarifocalNet (D19)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.037	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

1) The table presents FNs in the physical world under white-box and black-box attacks.
2) The light green area (first column) and light red area (second column) represent the adversarial attack methods and the aerial detectors employed for training adversarial backgrounds, while the light gray area (first row) denotes the aerial detectors used for testing.
3) The color in the table is associated with the average confidence score. Value closer to 0 is indicated by red color, reflecting better attack performance of the adversarial background, while value closer to 1 is indicated by blue color, signifying poorer attack performance of the adversarial background.

targeted objects, intending to misclassify them as small vehicles.

Performing adversarial attack detection with 19 different aerial detectors, we individually record the confidence scores

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TABLE IV
PHYSICAL TARGETED ATTACK RESULTS COMPARISON

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	**Average**
Clean	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
SSD	0.93	0.91	0.93	0.90	0.95	0.94	0.88	0.80	0.92	0.91	0.94	0.96	0.94	0.94	0.59	0.86	0.94	0.58	**0.88**
Faster R-CNN	0.84	0.70	0.92	0.93	0.53	0.97	0.96	0.90	0.94	0.95	0.96	0.83	0.81	0.74	0.94	0.89	0.97	0.53	**0.85**
Swin Transformer	1.00	1.00	1.00	0.99	0.99	1.00	0.99	1.00	1.00	1.00	1.00	1.00	0.98	1.00	0.00	1.00	1.00	1.00	**0.94**
YOLOv3	0.80	0.77	0.80	0.83	0.85	0.88	0.82	0.83	0.81	0.79	0.79	0.66	0.86	0.86	0.82	0.77	0.85	0.77	**0.81**
YOLOv5n	0.90	0.85	0.78	0.76	0.73	0.90	0.86	0.78	0.85	0.86	0.80	0.83	0.87	0.80	0.86	0.84	0.87	0.75	**0.83**
YOLOv5s	0.74	0.83	0.56	0.75	0.84	0.85	0.65	0.48	0.76	0.75	0.80	0.51	0.76	0.75	0.61	0.69	0.81	0.81	**0.72**
YOLOv5m	0.85	0.79	0.82	0.82	0.64	0.88	0.85	0.85	0.85	0.83	0.84	0.86	0.89	0.49	0.59	0.88	0.88	0.86	**0.80**
YOLOv5l	0.90	0.91	0.87	0.84	0.89	0.91	0.84	0.72	0.81	0.91	0.90	0.90	0.91	0.91	0.89	0.87	0.89	0.80	**0.87**
YOLOv5x	0.89	0.89	0.87	0.89	0.87	0.90	0.89	0.87	0.84	0.89	0.89	0.48	0.81	0.00	0.86	0.85	0.87	0.77	**0.80**
Cascade R-CNN	0.91	0.97	0.92	0.95	0.97	0.98	0.63	0.70	0.45	0.94	0.79	0.88	0.98	0.96	0.00	0.00	0.00	0.00	**0.67**
RetinaNet	0.88	0.89	0.91	0.93	0.96	0.91	0.79	0.82	0.80	0.89	0.85	0.83	0.94	0.83	0.89	0.91	0.95	0.87	**0.88**
Mask R-CNN	0.97	0.51	0.77	0.90	0.67	0.96	0.76	0.85	0.64	0.82	0.64	0.91	0.95	0.61	0.91	0.97	0.98	0.81	**0.81**
FreeAnchor	1.00	0.97	0.99	1.00	0.98	0.99	1.00	0.97	0.99	0.94	0.98	0.98	1.00	0.98	0.99	0.99	1.00	0.95	**0.98**
FSAF	0.91	0.93	0.86	0.91	0.91	0.93	0.68	0.82	0.79	0.78	0.00	0.75	0.84	0.71	0.90	0.94	0.94	0.67	**0.79**
RepPoints	0.86	0.88	0.92	0.89	0.89	0.93	0.75	0.94	0.85	0.80	0.87	0.93	0.92	0.91	0.86	0.94	0.92	0.83	**0.88**
TOOD	0.77	0.81	0.72	0.79	0.79	0.79	0.77	0.83	0.89	0.46	0.55	0.74	0.64	0.45	0.83	0.84	0.78	0.75	**0.73**
ATSS	0.66	0.71	0.66	0.68	0.59	0.75	0.65	0.63	0.64	0.56	0.67	0.50	0.69	0.63	0.58	0.67	0.74	0.64	**0.65**
FoveaBox	0.70	0.71	0.82	0.85	0.85	0.94	0.90	0.92	0.98	0.80	0.75	0.74	0.82	0.68	0.90	0.90	0.92	0.72	**0.83**
VarifocalNet	0.86	0.83	0.86	0.89	0.87	0.89	0.84	0.83	0.86	0.84	0.82	0.87	0.88	0.76	0.81	0.72	0.88	0.79	**0.84**

1) The table presents the highest confidence score of detecting an aircraft as a small-vehicle under white-box attacks in the physical world;
2) All results in the table are based on situations where no aircraft are detected;
3) Higher numerical values indicate a greater confidence in detecting a small-vehicle, highlighting the more significant impact of FPs on the performance of the aerial detector in the physical world.

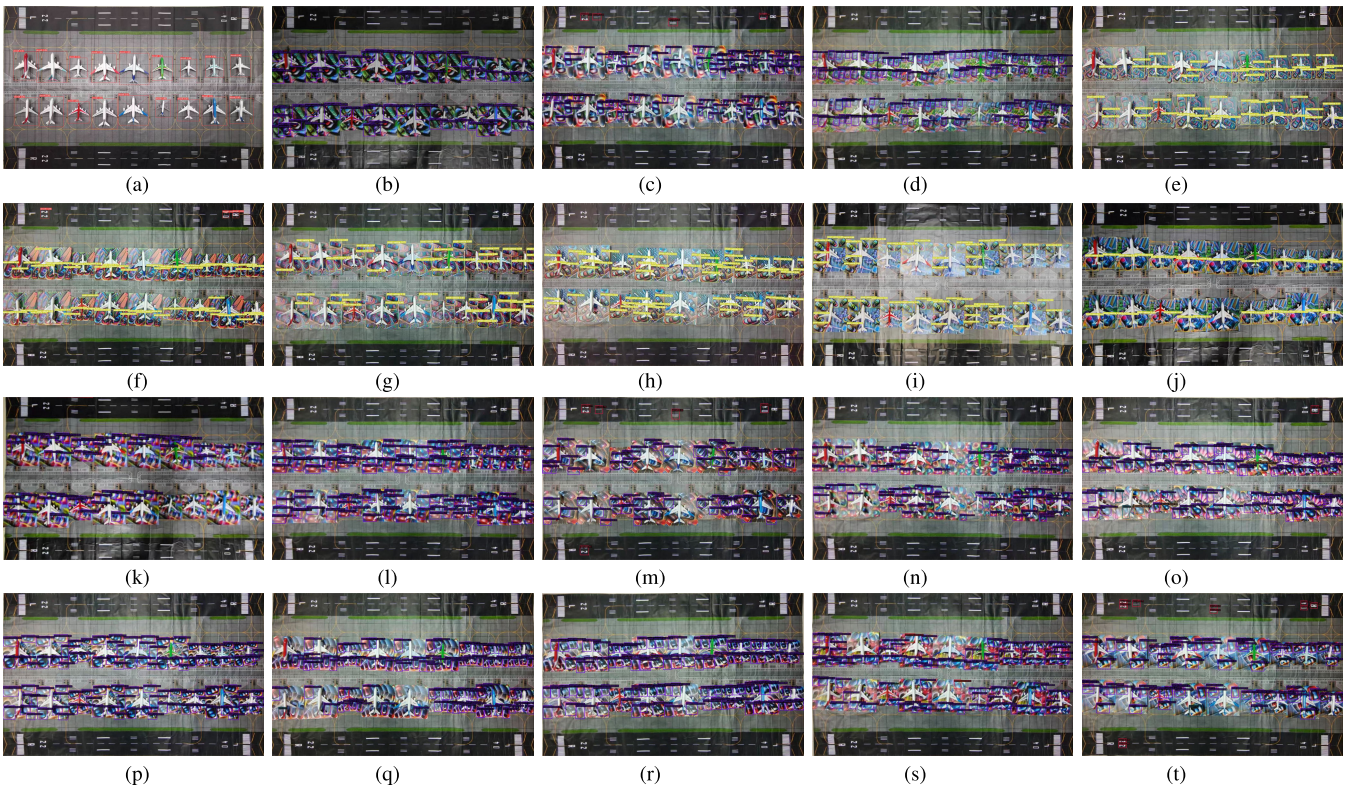


Fig. 8. Detection results of targeted adversarial backgrounds FPs in the physical world. (a) Clean. (b) SSD. (c) Faster R-CNN. (d) Swin Transformer. (e) YOLOv3. (f) YOLOv5n. (g) YOLOv5s. (h) YOLOv5m. (i) YOLOv5l. (j) YOLOv5x. (k) Cascade R-CNN. (l) RetinaNet. (m) Mask R-CNN. (n) FreeAnchor. (o) FSAF. (p) RepPoints. (q) TOOD. (r) ATSS. (s) FoveaBox. (t) VarifocalNet.

of 18 aircraft detected as small vehicles. The experimental results are documented in Table IV. Meanwhile, the detection results are shown in Fig. 8. Through an analysis of the results, the following conclusions can be drawn.

1) In all white-box attack scenarios, where the target model serves as the proxy model, our proposed BADEI method significantly reduces the detection performance of the aerial detectors, which implies that aircraft cannot be

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TABLE V
EFFECT OF HYPERPARAMETER SETTINGS ON THE EFFECTIVENESS OF TARGETED ATTACKS

β	γ	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	**Average**
0	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	0.1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
0.1	0	0.86	0.78	0.48	0.60	0.81	0.90	0.85	0.84	0.75	0.44	0.00	0.52	0.73	0.85	0.79	0.84	0.88	0.85	**0.71**
	0.1	0.87	0.90	0.80	0.89	0.84	0.86	0.85	0.78	0.82	0.87	0.87	0.58	0.80	0.62	0.70	0.79	0.87	0.81	**0.81**
	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
1	0	0.85	0.86	0.77	0.75	0.82	0.86	0.79	0.82	0.79	0.77	0.56	0.00	0.52	0.44	0.64	0.76	0.82	0.67	**0.69**
	0.1	0.85	0.79	0.81	0.80	0.70	0.82	0.76	0.83	0.81	0.81	0.84	0.78	0.80	0.61	0.73	0.84	0.85	0.80	**0.79**
	1	0.90	0.84	0.76	0.76	0.73	0.90	0.86	0.78	0.85	0.86	0.80	0.83	0.87	0.80	0.86	0.87	0.87	0.75	**0.83**
	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
10	0	0.82	0.83	0.82	0.74	0.69	0.88	0.81	0.83	0.83	0.77	0.80	0.80	0.81	0.74	0.55	0.85	0.85	0.85	**0.79**
	0.1	0.77	0.64	0.43	0.00	0.00	0.57	0.76	0.71	0.66	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.53	0.00	**0.28**
	1	0.42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.62	0.63	**0.09**
	10	0.71	0.53	0.00	0.00	0.00	0.76	0.78	0.72	0.74	0.00	0.00	0.00	0.00	0.00	0.65	0.72	0.84	0.81	**0.40**
	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
100	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	0.1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
	100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**

This table shows the confidence scores for identifying 18 aircraft as small-vehicles with different hyperparameter settings. The higher the confidence score, the higher the effectiveness of the targeted physical attacks.

TABLE VI
ABLATION STUDY ON LOSS MODULES FOR RECOGNIZING SPECIFIED CATEGORY TARGETS

L_{FN}	L_{TV}	L_{FP}	L_{Box}	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	**Average**
×	-	-	-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	×	×	×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	×	✓	×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	×	×	✓	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	✓	✓	×	0.85	0.86	0.77	0.75	0.82	0.86	0.79	0.82	0.79	0.77	0.56	0.00	0.52	0.44	0.64	0.76	0.82	0.67	**0.69**
✓	✓	×	✓	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	✓	✓	✓	0.90	0.84	0.76	0.76	0.73	0.90	0.86	0.78	0.85	0.86	0.80	0.83	0.87	0.80	0.86	0.87	0.87	0.75	**0.83**

This table shows the confidence scores for identifying 18 aircraft as small-vehicles.

detected. In addition, they are more likely to be identified as small-vehicles with high confidence scores.

2) FPs can almost perfectly identify aircraft as small vehicles, with the exception of the cascade R-CNN and ATSS models, where the confidence scores for detecting small vehicles are relatively low. However, the average confidence scores still reach 0.67 and 0.65, respectively. This indicates that meticulously designed FPs exhibit outstanding targeted attack effectiveness, making them more suitable for practical applications in the real world.

D. Hyperparameter Setting

Hyperparameters α, β, and γ are utilized to adjust the weight of different components of the loss. Specifically, to ensure a fair comparison with other methods, α is set to 1.5. By analyzing and discussing the parameters β and γ associated with targeted loss, we investigate the impact of adversarial backgrounds generated under different parameter settings on the effectiveness of physical attacks. The values of these two parameters are set to 0, 0.1, 1, 10, and 100, and the experimental results are presented in Table V. It is evident that different parameter settings for β and γ significantly influence the experimental outcomes. Notably, when β and γ are set to 1 individually, the confidence score of identifying the aircraft as small vehicles is the highest. It is important to highlight that all experiments in this article are conducted with α, β, and γ set to 1.5, 1, and 1, respectively.

E. Ablation Study

In this section, we discuss the impact of different loss components on physical attacks. The targeted loss function of our proposed BADEI method comprises four parts: L_{FN},

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TABLE VII
ABLATION STUDY ON LOSS MODULES FOR HIDING TARGETS

L_{FN}	L_{TV}	L_{FP}	L_{Box}	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	**Average**
×	-	-	-	0.91	0.89	0.90	0.89	0.90	0.88	0.88	0.90	0.90	0.90	0.90	0.90	0.92	0.91	0.85	0.91	0.90	0.91	**0.90**
✓	×	×	×	0.90	0.85	0.90	0.55	0.81	0.83	0.80	0.86	0.90	0.88	0.89	0.90	0.61	0.87	0.85	0.90	0.87	0.90	**0.84**
✓	✓	×	×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	×	✓	×	0.87	0.87	0.82	0.86	0.87	0.71	0.87	0.85	0.84	0.87	0.87	0.84	0.87	0.87	0.82	0.83	0.88	0.86	**0.85**
✓	×	×	✓	0.85	0.88	0.86	0.84	0.88	0.78	0.84	0.86	0.83	0.85	0.88	0.79	0.86	0.86	0.82	0.84	0.88	0.89	**0.85**
✓	✓	✓	×	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	✓	×	✓	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**
✓	✓	✓	✓	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**0.00**

This table shows the confidence scores for detecting 18 aircraft.

L_{TV}, L_{FP}, and L_{Box}. Ablation experiments are conducted in the physical world by training various adversarial backgrounds via the proxy model YOLOv5n. These backgrounds are printed and placed directly beneath the targets for detection. To evaluate the attack effectiveness of different loss modules, we compare the confidence scores of identifying the aircraft as small vehicles. Experimental results are presented in Table VI, highlighting the crucial role of L_{FP} in successfully misclassifying aircraft as small vehicles. It is worth noting that without L_{TV}, targeted attacks cannot be implemented even with L_{FP}. This is because L_{TV} is fundamental to the transition from digital attacks to physical attacks. In addition, the inclusion of bounding box constraints enables the detection of small vehicles with higher confidence scores.

Simultaneously, we record the confidence scores for detected aircraft, as shown in Table VII. It is evident that L_{TV} is similarly essential in the physical world and L_{FN} plays a pivotal role in capably concealing the targets. When L_{FN} is not applied as a constraint to the adversarial backgrounds, the average confidence score for detecting aircraft is 0.90. However, introducing L_{FN} as a constraint facilitates successful evasion of detection, while the other two loss modules L_{FP} and L_{Box} show no considerable impact on target concealment.

V. CONCLUSION AND DISCUSSION

This article proposes a dual-adversarial-induced approach to attack detectors by fooling background as targets, as well as fooling targets as background, by which the elaborated FNs and FPs achieve excellent performance in both the digital domain and the physical world for both nontargeted and targeted attacks. Prior studies on patch-based background attacks have often failed to make sufficient use of background information, resulting in potential target occlusion. Therefore, our proposed BADEI framework reaches significant attack effectiveness by implementing a seamless background attack. Besides, there is a notable disparity between the previous training strategy and the actual testing, in which to some extent diminishes the effectiveness of adversarial patches against certain aerial detectors. Therefore, we introduce the UTS, where adversarial backgrounds are placed directly below the targets during the adversarial background training process, achieving consistency with practical experiments. This strategy aims to minimize target occlusion while maximizing the utilization of background information to influence aerial detectors more effectively, which is crucial in enhancing the adversarial robustness and transferability of the generated backgrounds. Furthermore, given that existing adversarial attacks on aerial detectors typically focus on single-induction attacks, i.e., deceiving targets as background, we design D^{2}LF from the perspective of the loss function to craft FNs and FPs. The effectiveness of D^{2}LF has been validated in both nontargeted and targeted attacks. Finally, through extensive and rigorous experiments, the effectiveness of BADEI is confirmed.

The adversarial backgrounds successfully deceive aerial detectors via dual-induced error identification by misrepresenting background as targets, as well as misrepresenting targets as background. In addition, the proposed approach outperforms previous works in digital domain experiments and demonstrates promising attack effects in the dynamic physical environment, especially against more robust aerial detectors. However, there is still some room for improvement in the effectiveness of very few adversarial backgrounds for attacking certain detectors. Besides, this article experiments are conducted by printing off rectangular adversarial backgrounds from the printer and using the model aircraft. In the future, experiments over real-world targets would be considered. Meanwhile, we would like to further camouflage the adversarial background to improve the efficiency of attacks on detectors such as YOLOv5m, YOLOv5l, YOLOv5x, Cascade R-CNN, VarifocalNet, and others. In addition, the form of the adversarial attacks, as well as the location and shape of the adversarial background or adversarial patches, should be deeply explored.

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